Incident Response: Rapid Disk Exhaustion

DevOps Case Study: Log Flooding Analysis

Resolved & Documented

Incident Overview: Detected critical disk space depletion in the `/var/log` partition. Identified a rogue process generating excessive log data and captured forensic evidence for root cause analysis in a tool-restricted environment.

1. Crisis Detection

The `/var/log/messages` file was observed growing at an unsustainable rate of **~15 GB per hour**. Disk utilization reached ***92%***, risking a total system hang.

System Constraints:

- Environment: Minimalistic Linux installation.
- Unavailable Tools: `lsof`, `fuser`, `iotop`.
- Challenge: Real-time identification of a high-frequency write process without standard forensic utilities.

2. Forensic Investigation

Utilized the `top` utility to monitor process states and resource consumption. The culprit was identified by its high ***%CPU*** and frequent transitions into the ***D*** (Uninterruptible Sleep) state.

Identified Process: `log_generator` (PID 62)

3. Evidence Collection & Action

The requirement was to isolate the process details and extract the specific log pattern to the `/home/devops/excessive_log_process.txt` repository.

```
# Capturing process forensics
ps -p 62 -f > /home/devops/excessive_log_process.txt

# Extracting log signatures for debugging
tail -n 50 /var/log/messages >> /home/devops/
excessive_log_process.txt
```

4. Root Cause Hypothesis

The log pattern suggested a recursive loop in a testing script. By redirecting forensic output to a persistent file, the development team could debug the `log_generator` service while the ops team implemented log rotation to prevent future exhaustion.

Methods: /proc filesystem analysis, top monitoring, stream redirection.